

SPARQL

CQ.1 What are the modalities of APA (frequency, intensity) for physical activity and muscle building?

```

PREFIX      rdf:      <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX      owl:    <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX :    <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>

SELECT DISTINCT ?prop ?modalite WHERE {
:RenforcementMusculaire rdfs:subClassOf* ?parent .

?parent (rdfs:subClassOf owl:equivalentClass) ?anonClass .
?anonClass owl:intersectionOf ?list .
?list rdf:rest*/rdf:first ?element .

?element a owl:Restriction ; owl:onProperty ?prop ; owl:hasValue ?modalite .

FILTER(?prop IN (:aPourFrequence, :aPourIntensite))
}

```

SPARQL query:	
<pre> WHERE { :RenforcementMusculaire rdfs:subClassOf* ?parent . ?parent (rdfs:subClassOf owl:equivalentClass) ?anonClass . ?anonClass owl:intersectionOf ?list . ?list rdf:rest*/rdf:first ?element . ?element a owl:Restriction ; owl:onProperty ?prop ; owl:hasValue ?modalite . FILTER(?prop IN (:aPourFrequence, :aPourIntensite)) } </pre>	
prop	modalite
aPourFrequence	"par semaine"
aPourIntensite	Borg modéré

CQ2. What physical activities are prohibited in the context of inflammatory heart disease?

```
PREFIX      rdf:      <http://www.w3.org/1999/02/22-rdf-syntax-ns#>      PREFIX      owl:
<http://www.w3.org/2002/07/owl#>
PREFIX      rdfs:      <http://www.w3.org/2000/01/rdf-schema#>      PREFIX      xsd:
<http://www.w3.org/2001/XMLSchema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
```

```
SELECT DISTINCT ?activite WHERE {
:CardiopathiesInflammatoires rdfs:subClassOf* ?parent .
?parent (rdfs:subClassOf|owl:equivalentClass) [ owl:intersectionOf ?list ] .
?list rdf:rest*/rdf:first ?element .
?element a owl:Restriction ;
owl:onProperty :aPourContreIndication ; owl:someValuesFrom ?activite .
}
```

SPARQL query:

```
SELECT DISTINCT ?activite
WHERE {
:CardiopathiesInflammatoires rdfs:subClassOf* ?parent .

?parent (rdfs:subClassOf|owl:equivalentClass)
[ owl:intersectionOf ?list ] .

?list rdf:rest*/rdf:first ?element .

?element a owl:Restriction ;
owl:onProperty :aPourContreIndication ;
owl:someValuesFrom ?activite .
}
```

Sport
RenforcementMusculaire
ActiviteMixte
ActiviteEndurance

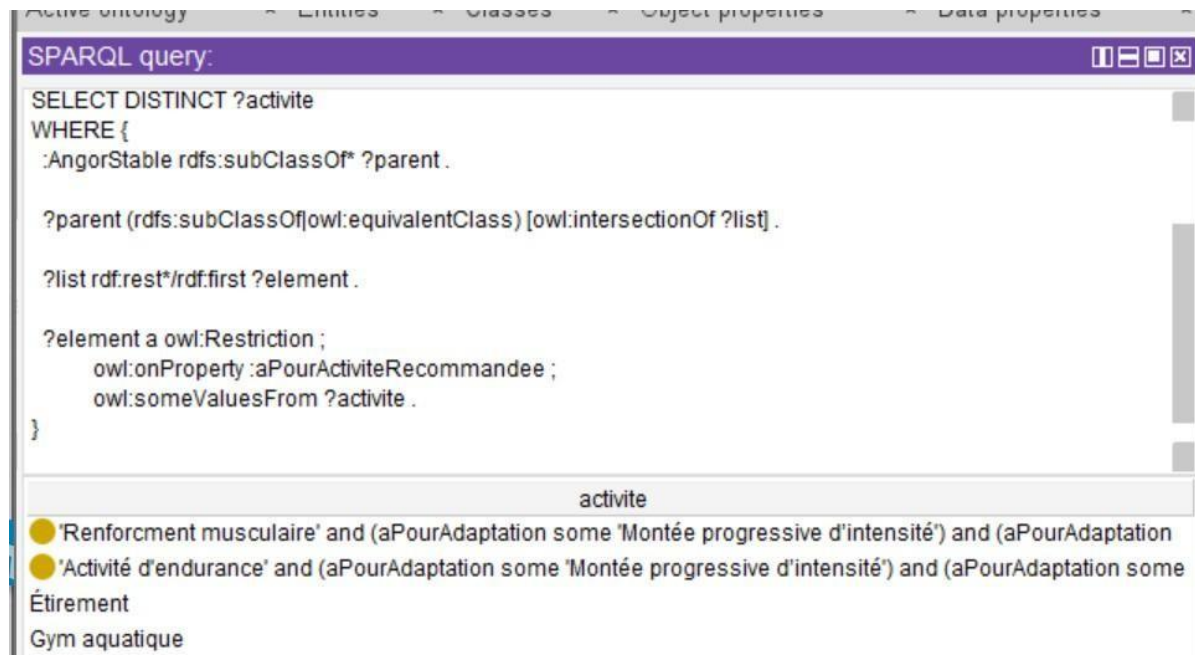
CQ3. What physical activities are recommended for stable angina pectoris?

```
PREFIX      rdf:      <http://www.w3.org/1999/02/22-rdf-syntax-ns#>      PREFIX      owl:
<http://www.w3.org/2002/07/owl#>
PREFIX      rdfs:      <http://www.w3.org/2000/01/rdf-schema#>      PREFIX      xsd:
<http://www.w3.org/2001/XMLSchema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
SELECT DISTINCT ?activite WHERE {
:AngorStable rdfs:subClassOf* ?parent .

?parent (rdfs:subClassOf[owl:equivalentClass] [owl:intersectionOf ?list] .

?list rdf:rest*/rdf:first ?element .

?element a owl:Restriction ;
owl:onProperty :aPourActiviteRecommandee ; owl:someValuesFrom ?activite .
}
```



CQ4. What facilitator should be offered to a patient who does not have a gym near their home?

```
PREFIX      rdf:      <http://www.w3.org/1999/02/22-rdf-syntax-ns#>      PREFIX      owl:
<http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>

SELECT DISTINCT ?levier WHERE {
:AbsenceGymnase rdfs:subClassOf* ?parent .
?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
?anon owl:intersectionOf ?list .
?list rdf:rest*/rdf:first ?element .
?element a owl:Restriction ; owl:onProperty :aPourLevier ;
owl:someValuesFrom ?unionClass .
?unionClass owl:unionOf ?uList .
?uList rdf:rest*/rdf:first ?levier .
}
```

SPARQL query: 

```
SELECT DISTINCT ?levier
WHERE {
:AbsenceGymnase rdfs:subClassOf* ?parent .
?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
?anon owl:intersectionOf ?list .
?list rdf:rest*/rdf:first ?element .
?element a owl:Restriction ;
      owl:onProperty :aPourLevier ;
      owl:someValuesFrom ?unionClass .
?unionClass owl:unionOf ?uList .
?uList rdf:rest*/rdf:first ?levier .
}
```

levier
Activité à domicile
Activité sans matériel
Marche autour du domicile
Micro séance 10 min

CQ5. What contraindications to physical activity are associated with acute coronary syndrome in a patient also with hypertension and a stent?

```

PREFIX      rdf:      <http://www.w3.org/1999/02/22-rdf-syntax-ns#>      PREFIX      owl:
<http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>

SELECT DISTINCT ?profil ?contreIndication WHERE {
VALUES ?profil {
:SyndromeCoronarienAigu
:Hypertension_arterielle
}

?profil rdfs:subClassOf* ?parent .
?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
?anon owl:intersectionOf ?list .
?list rdf:rest*/rdf:first ?restriction .

?restriction a owl:Restriction ;
owl:onProperty :aPourContreIndication ; owl:someValuesFrom ?contreIndication .
}

```

SPARQL query: ⏏ ⌵ ⌶ ⌵	
<pre> :Hypertension_arterielle } ?profil rdfs:subClassOf* ?parent . ?parent (rdfs:subClassOf owl:equivalentClass) ?anon . ?anon owl:intersectionOf ?list . ?list rdf:rest*/rdf:first ?restriction . ?restriction a owl:Restriction ; owl:onProperty :aPourContreIndication ; owl:someValuesFrom ?contreIndication . } </pre>	
profil	contreIndication
Syndrome coronarien aigu	Activité physique
Hypertension artérielle	Travail en isométrie
Hypertension artérielle	Musculation lourde

CQ6. What equipment is needed to perform a leg abduction exercise?

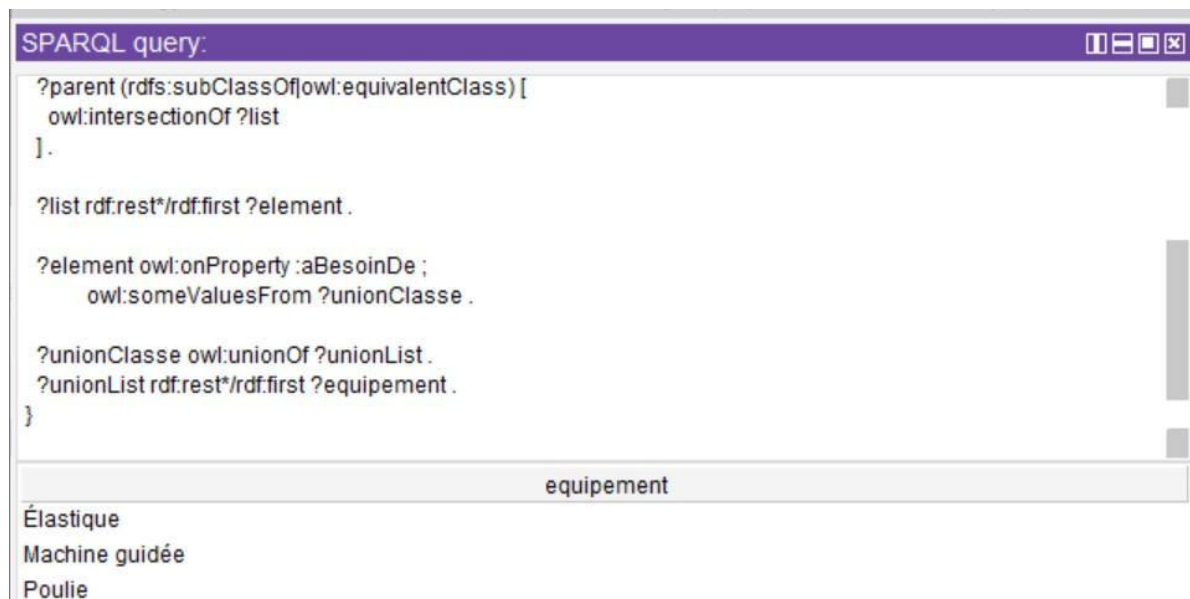
```
PREFIX      rdf:      <http://www.w3.org/1999/02/22-rdf-syntax-ns#>      PREFIX      owl:
<http://www.w3.org/2002/07/owl#>
PREFIX      rdfs:      <http://www.w3.org/2000/01/rdf-schema#>      PREFIX      xsd:
<http://www.w3.org/2001/XMLSchema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
SELECT DISTINCT ?equipment WHERE {
:AbductionDeJambe rdfs:subClassOf* ?parent .

?parent (rdfs:subClassOf|owl:equivalentClass) [ owl:intersectionOf ?list
] .

?list rdf:rest*/rdf:first ?element .

?element owl:onProperty :aBesoinDe ; owl:someValuesFrom ?unionClasse .

?unionClasse owl:unionOf ?unionList .
?unionList rdf:rest*/rdf:first ?equipment .
}
```



CQ7. Which physical activities require adaptation in the context of functional coronary artery disease?

```

PREFIX      rdf:      <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX      owl:    <http://www.w3.org/2002/07/owl#>
PREFIX      rdfs:      <http://www.w3.org/2000/01/rdf-schema#>
PREFIX      xsd:      <http://www.w3.org/2001/XMLSchema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
SELECT DISTINCT ?actFiller ?adaptation WHERE {
:CoronaropathieFonctionnelle rdfs:subClassOf* ?parent .
?parent (rdfs:subClassOf|owl:equivalentClass) ?ce .
?ce owl:intersectionOf ?list .
?list rdf:rest*/rdf:first ?elemAct .

?elemAct a owl:Restriction ;
owl:onProperty :aPourActiviteRecommandee ; owl:someValuesFrom ?actFiller .

?actFiller owl:intersectionOf ?actList2 .
?actList2 rdf:rest*/rdf:first ?adRest .

?adRest a owl:Restriction ;
owl:onProperty :aPourAdaptation .
{
?adRest owl:someValuesFrom ?adaptation .
FILTER NOT EXISTS { ?adaptation owl:unionOf ?u }
} UNION
{
?adRest owl:someValuesFrom ?uClass .
?uClass owl:unionOf ?u .
?u rdf:rest*/rdf:first ?adaptation .
}
}
ORDER BY ?actFiller ?adaptation

```

SPARQL query:	
<pre> { ?adRest owl:someValuesFrom ?adaptation . FILTER NOT EXISTS { ?adaptation owl:unionOf ?u } } UNION { ?adRest owl:someValuesFrom ?uClass . ?uClass owl:unionOf ?u . ?u rdf:rest*/rdf:first ?adaptation . } } ORDER BY ?actFiller ?adaptation </pre>	
actFiller	adaptation
● 'Renforcement musculaire'	and (aPourAdaptation some Montée progressive d'intensité
● 'Renforcement musculaire'	and (aPourAdaptation some Réduction de charge
● 'Activité d'endurance'	and (aPourAdaptation some 'Mor Borg faible
● 'Activité d'endurance'	and (aPourAdaptation some 'Mor Borg modéré
● 'Activité d'endurance'	and (aPourAdaptation some 'Mor Montée progressive d'intensité

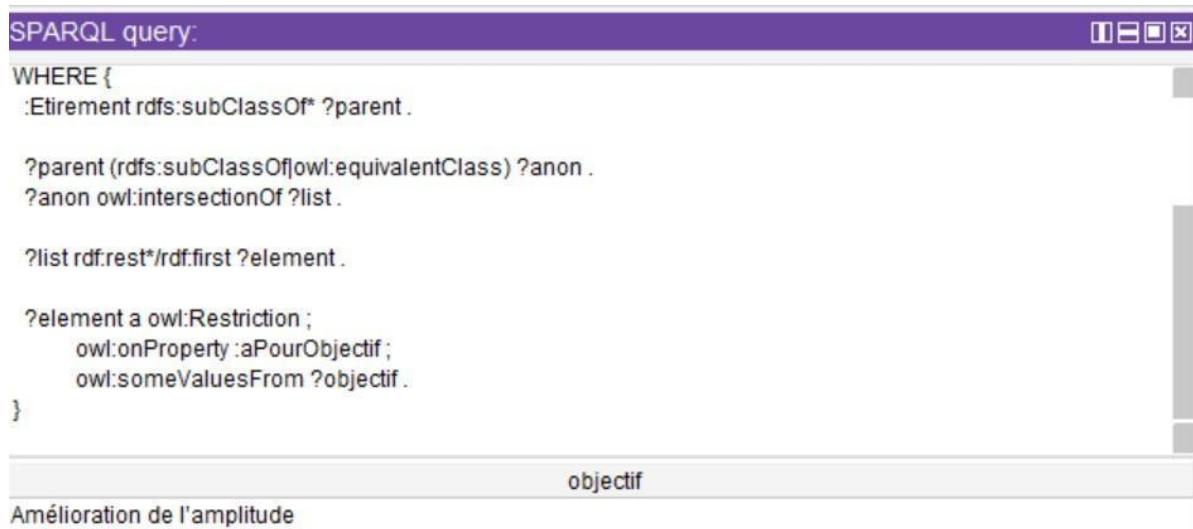
CQ8. What are the objectives associated with stretching exercises?

```
PREFIX      rdf:      <http://www.w3.org/1999/02/22-rdf-syntax-ns#>      PREFIX      owl:
<http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
SELECT DISTINCT ?objectif WHERE {
:Etirement rdfs:subClassOf* ?parent .

?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
?anon owl:intersectionOf ?list .

?list rdf:rest*/rdf:first ?element .

?element a owl:Restriction ;
  owl:onProperty :aPourObjectif ;
  owl:someValuesFrom ?objectif .
}
```



CQ9. What physical exercises were prescribed for X?

```
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
SELECT ?prescription ?date ?activite
WHERE {
  ?prescription :concerne :Patient1 .
  ?prescription :aPourDate ?date .
  ?prescription :contient ?activite .
}
```

SPARQL query:		
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>		
SELECT ?prescription ?date ?activite		
WHERE {		
?prescription :concerne :Patient1 .		
?prescription :aPourDate ?date .		
?prescription :contient ?activite .		
}		
prescription	date	activite
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	ActivitéEndurancePatient1
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	Renforcement musculaire Patient 1
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	Etirement patient 1
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	Sport_de_combat_Patient_1